

RFM Model Customer Segmentation Based on Hierarchical Approach Using FCA

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ARTICLE INFO

Keywords:

Customer Segmentation
Formal Concept Analysis
Hierarchical concept
RFM model
Clustering

ABSTRACT

Nowadays, every business focuses on customer relationship management (CRM) to deliver their customers better services and to establish a competitive advantage over their competitors. Significantly, customer insights with solid customer relationships improve customer retention and satisfaction, thereby contributing to profit. Thus, customer segmentation based on cluster analysis is critical to customer identification in CRM. In addition, it can identify the potential customers and their needs to be matched with marketing strategies. However, unfortunately, this approach led to a gap between the marketing persons who care about the business implications and clustering output with the data science complexity barrier. Moreover, most clustering methodologies give only groups or segments, such that customers of each group have similar features without customer data relevance. Thus, this work sought to address these concerns by using a hierarchical approach. This research proposes a new effective clustering algorithm by combining the RFM (Recency, Frequency, and Monetary) model with FCA. This new methodology uses the advantages of FCA in building the knowledge representation; therefore, the obtained construction contains both implicit and explicit knowledge. Explicit knowledge shows information represented in the hierarchical structure model, while implicit knowledge is embedded in the structure with its implication properties. Thus, the knowledge structure from FCA reveals relationships among data points in an easily understood manner. The proposed model was evaluated and compared with K-means clustering and hierarchical clustering using the online retail II dataset from the UCI Machine Learning Repository. The proposed method provides enough and appropriate information for marketers to perceive the value of the clustering results for creating practical marketing strategies in real-world business by offering the marketers both customer segmentation and the relationships in customer data at the same time.

1. Introduction

STP (Segmentation Targeting Positioning) marketing is a core marketing approach for creating superior customer value and supporting the development of products and services (Munusamy and Murugesan (2020); Gupta, Justy, Kamboj, Kumar and Kristoffersen (2021)). The first step mainly addresses determining important characteristics to differentiate each market segment, leading to market targeting and product positioning. In competitive markets, businesses need to understand their customers to generate suitable matched marketing strategies. However, it is difficult to understand a massive number of customers clearly (Deng and Gao (2020); Chen, Zhang, Chu and Yan (2019b)). Because of this, before companies can apply marketing strategies to their customers, they often use customer segmentation (also called market segmentation) to categorize the customers. Customer segmentation divides customers into groups according to their similarities in needs, characteristics, or behaviors to maintain customer relationships and increase profit. Moreover, customer segmentation is a tool of customer identification that is the primary process of customer relationship management (CRM) (Deng and Gao (2020); Dolnicar, Grün and Leisch (2018a)). Notably, customer segmentation supports businesses in enhancing customer retention and loyalty and helps identify the value of a customer (Ballestar, Grau-Cardes and Sainz (2018); Wu and Liu (2020); Nandapala and Jayasena (2020)).

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There are several customer segmentation methods, but most of them are based on customers' behavioral, psychographic, geographic, and demographic information. Segmentation algorithms can be divided into four main groups. They include association algorithms (e.g., Apriori, FP growth, ECLAT partition), clustering (e.g., hierarchical clustering, K-means clustering, fuzzy C Means clustering, density-based clustering, affinity propagation clustering), classification algorithms (e.g., KNN, Naïve Bayes, SVM, C4.5, Decision Tree) and regression algorithms (e.g., Logistic regression) (Tsipitsis and Chorianopoulos (2011)). Clustering is an unsupervised learning approach and can be divided into two categories: hard clustering and soft clustering (Singh and Srivastava (2020)). In hard clustering, each object is assigned to only one cluster, while soft clustering can also be done in an overlapping manner in which an object can be part of more than one cluster. After we have generated a partition using a clustering algorithm, we need to evaluate the validity or quality of this partition. If several partitions are generated (e.g., with different numbers of clusters), we need ways to compare them before we conclude the clustering result. There are three broad categories for cluster validity: internal, external, and relative indices (Tsipitsis and Chorianopoulos (2011)). The main approaches to assess these include graphical representations and internal indices. The silhouette plot is an advanced graphical representation that provides visual information about the quality of the partition. Internal indices measure a partition's "intrinsic" quality (how well-separated the clusters are). We have seen that the mean silhouette value can be used as an internal index. There exist many other internal indices. The Calinski–Harabasz, the Davies–Bouldin, and the Dunn indices are the three that are most widely used (Gagolewski, Bartoszek and Cena (2021)). Fuzzy Clustering is an example of soft clustering, and cluster validity indices have also been defined for this approach, one such index being the fuzzy silhouette index.

The K-means clustering algorithm is the most popular and is commonly used (Fränti and Sieranoja (2018); Khalili-Damghani, Abdi and Abolmakarem (2018)). However, it is sensitive to cluster centroid initialization and outliers, both impacting the eventual results (Deng and Gao (2020); Meng, Liu, Wu, Zhou, Xu and Wu (2020)). Unlike K-means, hierarchical clustering can produce a robust result without assigning initial values. Furthermore, the hierarchical clustering method is the most intuitive way of grouping data (Dolnicar, Grün and Leisch (2018b)). It allows a human to track the task of dividing a set of n observations (customers) into k clusters (segments) represented in a hierarchy represented graphically in a dendrogram. In contrast, the result of K-means is unstructured non-overlapping clusters; therefore, the hierarchical clustering result is more interpretable and informative for marketers.

Marketers need implicit and explicit knowledge to create practical marketing strategies in real-world business. However, hierarchical clustering gives only groups or segments of customers in a hierarchy lacking some knowledge in the form of implications such as the relevance of customer data. To address this problem, this current study proposes a new effective clustering algorithm using the advantages of the RFM model and formal concept analysis (FCA) to build knowledge representation and segmentation. The obtained construction contains both implicit and explicit knowledge. Explicit knowledge shows information represented in the hierarchical structure model, while implicit knowledge is embedded in the structure with its implication properties. Thus, the knowledge structure from FCA reveals relationships among data points in an easily understood manner. Above all, it is emphasized that a customer data representation based on FCA can be used to discover latent issues in customer information.

This new method's clustering was compared with the most popular method, K-means clustering, and hierarchical clustering, using the online retail II dataset from the UCI Machine Learning Repository. After the segmentation, we devised marketing strategies for each customer segment for business implication and interpretation. Finally, conclusions are shared.

2. Background

2.1. Customer segmentation

Customer segmentation is the process of dividing the customer group into subgroups according to similarities (Deng and Gao (2020); Choi, Choi, Yoon and Joung (2020); Singh and Mittal (2021); Zeybek (2018)). A simple customer segmentation approach includes geographic, demographic, psychographic, and behavioral segmentation. Behavioral segmentation, being the most commonly used method, includes the following main steps: 1) business understanding and design of the segmentation process; 2) data understanding, preparation, and enrichment; 3) identification of the segments with cluster modeling; 4) evaluation and profiling of the revealed segments; and 5) deployment of the segmentation solution, design, and delivery of the differentiated strategies (Tsipitsis and Chorianopoulos (2011));

Chorianopoulos (2016)).

Segmentation can use several alternative algorithms, including an association algorithm, clustering algorithm, classification algorithm, or regression algorithm. Among these, clustering is the most precise and effective method for customer segmentation (Tsiptsis and Chorianopoulos (2011)).

Customer segmentation with a clustering algorithm uses machine learning to classify customer data based on similarity. All clustering algorithms aim to minimize the distances within clusters and maximize the distances between them. Moreover, the different clustering algorithms and outcomes are related to the objective (cost) function to measure the quality of clustering, the underlying structure assumed, the similarity measure, and how to consider the number of clusters.

Above all, most clustering techniques give only groups or segments without customer data relevance, whereas the marketing persons need an effective customer segmentation that gives appropriate knowledge with an easily understood format.

2.2. RFM model

RFM model is one of the most prevalent behavioral segmentations (Peker, Kocyigit and Eren (2017); Khalili-Damghani et al. (2018); Wang, Tsai and Ciou (2020); Hosseini, Abdolvand and Harandi (2022)). This model groups existing customers with their recency, frequency, and monetary values and does not focus on attracting new customers but identifies the best customers to perform targeted marketing. Recency is the number of days since the last purchase. Frequency refers to the total number of purchases. Meanwhile, monetary is the total purchase value during a specific period (Dedi, Dzulhaq, Sari, Ramdhan, Tullah and Sutarman (2019); Sokol and Holý (2021)).

The RFM scoring process is carried out to demonstrate RFM analysis using the quintile method. The customers are split into quintiles (five equal groups), and each customer is given a score based on which quintile he belongs to (Christy, Umamakeswari, Priyatharsini and Neyaa (2021)). Accordingly, the scores assigned in the first quintile with the highest values for frequency (F) is 5, while the other quintiles are scored with 4, 3, 2, and 1 in rank order. This process is also undertaken for monetary (M). In contrast, the last quintile with the smallest recency (R) values is coded as 5. Next, we sequentially give the other quintiles based on recency values with the scores 4, 3, 2, and 1. Finally, all the customers are ranked using R, F, and M values. Accordingly, the best customer group is 5-5-5, and the worst is 1-1-1.

However, the other approaches to the RFM model use the actual values of each RFM factor as the segmentation variables. The original RFM values are determined using various clustering techniques with required data preprocessing. Especially, K-means is the most popular and most commonly used method (Ernawati, Baharin and Kasmin (2021)). Unfortunately, it is sensitive to cluster centroid initialization and outliers, so these affect the clustering results (Christy et al. (2021)). In addition, it gives only complete separation of customer groups without customer data relationships. Therefore, this study sought to create a practical approach to address these problems in using the RFM model, whose advantage is using a very small number of variables (only three variables), with clustering based on FCA.

2.3. Formal concept analysis

Formal concept analysis (FCA) is a mathematical theory of concept formation based on lattice theory, applied in classification and concept discovery to organize information and discover relationships (Ganter and Wille (2012); Wille (2009)). Moreover, FCA follows a human-centered approach and supports exploration operations through the concept lattice to organize information and discover relationships embedded in the binary relations between a pair of sets (called objects and attributes, respectively). A node in the concept lattice is an objects/attributes pair, called a (formal) concept. A concept consists of the extent (all objects belonging to the concept) and the intent (attributes describing the concept).

The core knowledge of the FCA approach is based on a simple data representation: a binary table called a formal context that is transformed into a mathematical structure called a concept lattice (Castellanos, Cigarrán and García-Serrano (2017)). $\mathbb{K} := (G, M, I)$ is a formal context in which G represents a set of objects, M represents a set of attributes and $I \subseteq G \times M$ represents a set of is-a or has-a relationships between G and M , defined by gIm , which is read as the object g has the attribute m . I is the incidence relation of the context (G, M, I) .

From this formal context, a set of formal concepts can be generated. To define a formal concept, the following derivation operations are needed. For any subsets A and B , $A \subseteq G$ and $B \subseteq M$:

$$\begin{aligned} A \mapsto A^I &:= \{m \in M \mid gIm \quad \forall g \in A\} \\ B \mapsto B^I &:= \{g \in G \mid gIm \quad \forall m \in B\} \end{aligned}$$

Thus, a formal concept is a pair (A, B) where $A \subseteq G$ is a set of objects (the extent of the formal concept), and $B \subseteq M$ is a set of attributes (the intent of the formal concept), which has the following properties:

- If an object a in A is tagged with an attribute b , then b must be included in B (i.e., $B = A^I$ the intent of the formal concept includes all the attributes shared by the objects in the extent).
- Conversely, if an object a is tagged with all the attributes in B , then a must be included in A (i.e., $A = B^I$: the extent of the formal concept includes all those objects filtered out by the intent).

Formal concepts can be ordered according to their extents by applying the partial order relationships (Benavent, Castellanos, de Ves, García-Serrano and Cigarrán (2019)) in equation (1), where a formal concept (A, B) with extent A is considered a sub-concept of another formal concept (C, D) with an extent C when the objects in A are contained into the objects in C .

$$(A, B) \leq (C, D) \Leftrightarrow (A \subseteq C \Leftrightarrow D \subseteq B). \quad (1)$$

The organization that results from this order relationship can be proven to be a lattice, a concept lattice, associated with the formal context denoted by $\mathfrak{B}(G, M, I)$. Since concept lattices are ordered sets, they can be naturally displayed in Hasse diagrams.

Frequency (support) is one of the most popular measures in the theory of pattern mining (Kuznetsov and Makhalova (2018)). Frequency arises from the assumption that the most "interesting" concepts are frequent ones:

$$\text{supp}(A, B) = \frac{|A|}{|G|} \quad (2)$$

The support provides an efficient level-wise algorithm of semilattice computing:

$$B_1 \subset B_2 \rightarrow \text{supp}(B_1) \geq \text{supp}(B_2). \quad (3)$$

In this study, we say that a set of attributes is frequent if its support exceeds a certain threshold. Thus, the frequency of an attribute set means that it is frequent.

Practically, clustering based on the FCA method contains four steps (Zhang, Zhao and Yan (2018)). First, the datasets are pretreated, and feature items are extracted from data elements. Then, a formal context is constructed by taking data elements as objects and feature items as attributes. Next, the concept of formal contexts is extracted, and the concept lattice of the formal contexts is constructed. Finally, a Hasse diagram as a highly informative visualization can be generated.

It is emphasized that the FCA provides a well-defined mathematical framework to discover implicit and explicit knowledge in an easily understood format by using formal context and a Hasse diagram that clearly represents the concepts' generalization/specialization relationships (Ganter and Wille (2012)). In addition, FCA can construct an informative concept hierarchy providing valuable information on various specific domains. Therefore, we propose performing RFM model customer segmentation based on FCA.

2.4. Association rule mining

An association rule is just a statement about the conditional sample probability (called confidence) of an event w.r.t. another one, together with the statement of the joint sample probability of the two events (called support), where both events are described in terms of attribute sets. In FCA terms, for two subsets of attributes (called itemsets in data

mining) Y_1 and $Y_2 \subseteq M$, the association rule $Y_1 \rightarrow Y_2$ has support $\frac{|(Y_1 \cup Y_2)'|}{|G|}$ and confidence $\frac{|Y_2'|}{|Y_1'|}$. The rule

$Y_1 \rightarrow Y_2$ is called frequent if $\text{supp}(Y_1 \rightarrow Y_2) \geq \text{minsupp}$ for some threshold minsupp . In FCA, association rules were known under the name of partial implications (Kuznetsov and Poelmans (2013)). This work applies the obtained FCA knowledge structure to extract implication rules to discover knowledge relationships in customer segmentation.

Table 1
Study of RFM-based Customer Segmentation

Resource	Dataset	Variables	Clustering techniques
Anitha and Patil (2022)	Online retail Chen et al. (2012)	RFM	K-means
Hosseini et al. (2022)	Real-time retail		
Christy et al. (2021)	Internet Banking	RFM	K-means and DBSCAN
	Online retail Chen et al. (2012)	RFM	K-means, RM K-means, and Fuzzy C-means
Frasquet et al. (2021)	Offline and Online retail	RFM + Sex + Age	Latent Class Analysis (LCA)
Rahim et al. (2021)	Online retail Chen et al. (2012)	RFM	K-means
Zhou et al. (2021)	Online retail	RFMT	Hierarchical
Shokouhyar et al. (2020)	Retail	RFM	K-means
Monalisa et al. (2019)	Distribution and retail	RFM	Fuzzy C-means
Nakano and Kondo (2018)	Multichannel Retail	RFM	Latent Class Analysis (LCA)

3. Related work

RFM model-based customer segmentation has been demonstrated using various clustering techniques, briefly summarized in Table 1. Most of these studies have applied K-means clustering to explicit customer segmentation. K-means is popular in the application areas because it is simple to understand, interpret, and apply. For example, Chen et al. applied K-means clustering and decision tree induction to segment customers from the online retail dataset of customer transactions in the UCI repository (Chen et al. (2012)). In the same way, many studies have deployed dataset segmentation by using the K-means algorithm (Anitha and Patil (2022); Christy et al. (2021); Chen et al. (2012); Anitha and Patil (2022)). These studies have shown that data preprocessing is a crucial and time-consuming process before segmentation, especially before K-means clustering. For this reason, data preprocessing needs to remove outliers, scale the ranges of data, and solve any Long Tails problems by data transformation (Tavakoli, Molavi, Masoumi, Mobini, Etemad and Rahmani (2018)). In addition, we found that the number of segments chosen was between 2 and 10 groups. The number of segments applied should not be too large because it will make it difficult for the marketing analyst to interpret and design marketing strategies for selected customer segments. The Silhouette index is the most widely used validation index. The online retail dataset of customer transactions introduced by Chen et al. (2012) and placed in the UCI repository is the most often used. Therefore, this work uses this dataset for benchmarking and comparing the proposed model with K-means clustering and hierarchical clustering.

Hierarchical clustering methods are the most intuitive way of grouping data because they mimic how a human would approach the task of dividing a set of observations (customers) into k groups (segments). Moreover, this is one of the most frequently used methods and can produce a robust result. Unlike other unsupervised methods, such as K-means, for hierarchical clustering it is not necessary to assign any initial values (Zhou et al. (2021)). Zhou et al. used hierarchical clustering for customer segmentation by web content mining. They used Calinski-Harabasz index and Davies-Doublin index to determine the optimal cluster number. In addition, Cheng et al. developed an approach to analyze customer behavior in mobile app usage and identify associations between functions by combining three data mining techniques: RFM analysis, link analysis based on graph theory, and association rule learning (Chen, Zhang and Zhao (2017)). Their proposed approach can be used to visualize and gain the advantages of these associations. Moreover, it provides insights into customer behaviors and function usage preferences. Remarkably, the prior work based on hierarchical clustering shows that segments and relationships are important and valuable knowledge for analyzing customer behaviors. Thus, this work proposes customer segmentation based on a hierarchical structure using FCA.

In (Lei, Yan, Han and Jiang (2018)), association rules were extracted from concept lattices generated by FCA. This study proposed a feasible and effective method based on concept lattice and attribute analysis to reduce the number of association rules. Furthermore, they controlled the number of concepts by using rough attribute values and improved the concept's quality. Above all, this study emphasized that association rules from concept lattices can provide latent and valuable knowledge.

However, in practice, a concept lattice has many formal concepts that lead to the high complexity of conceptual clustering. In (Nguyen, Tran, Quan, Nguyen and Le (2019)), Nguyen et al. introduced a framework named MarCHGen (Malware Concept Hierarchy Generation) to generate a malware concept hierarchy. In this framework, they established

Table 2

Variables in the customer transaction dataset (1,067,371 instances).

Variable name	Data type	Description; typical values and meanings
Invoice	Nominal	Invoice number; A 6-digit integral number uniquely assigned to each transaction. If this code starts with the letter 'c', it indicates a cancellation.
StockCode	Nominal	Product (item) code; A 5-digit integral number uniquely assigned to each distinct product.
Description	Nominal	Product (item) name;
Quantity	Numeric	The quantities of each product (item) per transaction.
InvoiceDate	Numeric	Invoice date and time; The day and time when a transaction was generated.
Price	Numeric	Unit price; Product price per unit in sterling (£).
Customer ID	Nominal	Customer number; A 5-digit integral number uniquely assigned to each customer.
Country	Nominal	Country name; The name of the country where a customer resides.

frequent concepts on a concept lattice. A frequent concept has object sets larger than a defined threshold. Furthermore, only the frequent concepts are kept on the lattice. Therefore, a frequent lattice will be pruned to be less complex than the original lattice.

Above all, hierarchical clustering gives only groups or segments of customers in a hierarchy lacking some knowledge in the form of implications such as the relevance of customer data. To counter this problem, this research proposes a new effective clustering algorithm using the advantages of the RFM model and formal concept analysis (FCA) to build knowledge representation and segmentation. Moreover, this appears to be the first time that the RFM model has been combined with the FCA.

4. Research methodology

The main contribution of this study is in applying FCA to build a knowledge structure for customer segmentation. FCA is ideal for identifying groups of customers with certain common properties (or features). The advantages of the presented structure are the discovery of explicit and implicit knowledge. The explicit knowledge is derived from identifying the co-appearance of attributes, while the implicit knowledge stems from the implication rules in relationships inside the concept lattice structure generated by the FCA. This section describes the process of building a hierarchical structure using FCA and its knowledge acquisition. Moreover, the traditional clustering approaches, K-means and agglomerative hierarchical clustering are compared with the proposed approach.

The proposed methodology can be divided into three main steps, as shown in Figure 1. The first step involves data exploration and preparation, which is critical for eventual accurate insights. Afterwards, the FCA knowledge structure provides explicit and implicit knowledge in customer segmentation and the implication relationships, respectively. The final step focuses on deploying the acquired knowledge in the customer segmentation and the discovered knowledge in each segment. The corresponding details are explained in the subsections that follow.

4.1. Exploratory Data Analysis and Preprocessing

Exploratory data analysis (EDA) is the primary data exploration to extract and understand the data patterns by using statistics and graphical representations. In this research, we use the online retail II dataset used in (Chen et al. (2012)), (Chen, Guo and Li (2019a)), (Christy et al. (2021)), and (Rahim et al. (2021)) from the UCI Machine Learning Repository. This dataset contains all the transactions over two years, occurring for a UK-based and registered, non-store online retail, from 1 December 2009 to 9 December 2011. The company mainly sells unique all-occasion giftware. Many customers of the company are wholesalers. The customer transaction dataset contains 1,067,371 records with eight variables, as shown in Table 2.

Practically, we identify the unique customers, details of data, inconsistent data, incomplete data (or missing values), and noisy data. We manipulate to prepare the dataset with cleaning data such as removing missing values and canceled/zero/negative transactions. Afterwards, the obtained dataset must be preprocessed for the required RFM model-based clustering analysis. The main steps and relevant tasks involved in the data preparation are as follows:

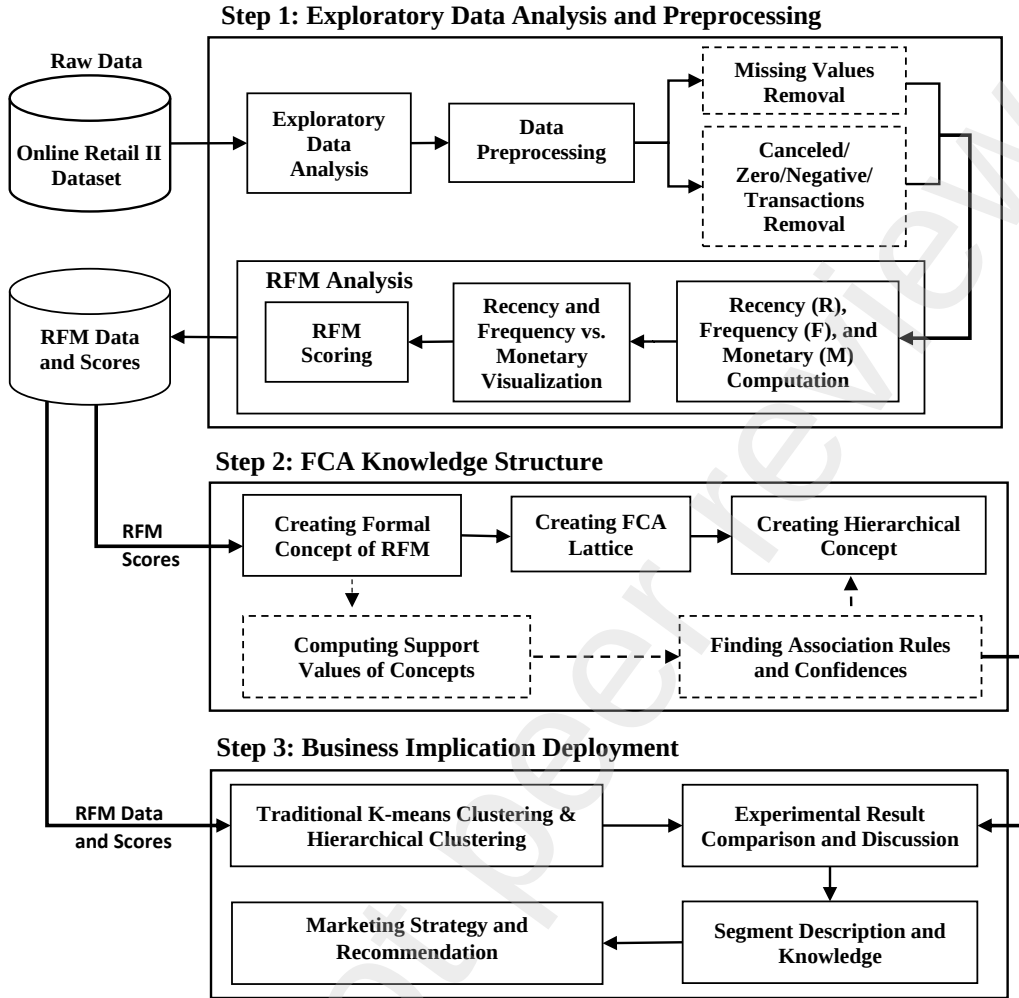


Figure 1: Tasks and techniques of the proposed approach.

1. Select appropriate variables of interest from the given dataset. In this work, the following five variables have been chosen: Invoice, Quantity, InvoiceDate, Price, and Customer ID.
2. Delete transactions with null values.
3. Delete invalid data of quantities and price variables less than or equal to zero.

After this data preparation, there were 805,620 transactions remaining in the target dataset. Next, the RFM analysis concept, which is a popular and significant customer segmentation in the retail industry, will be used to reduce the data before generating a knowledge structure. The recency of transactions, frequency, and amount the customers spent are determined to create RFM values. Moreover, recency, frequency, and monetary values are visualized to gain knowledge and understanding of these data. Next, the day of the last purchase was decided as the reference date to calculate recency. At last, each customer's RFM score was created.

To calculate the RFM score, we can see that all the quantities calculated here (recency, frequency, and monetary) have different ranges. All the customers are ranked by considering their recency, frequency, and monetary values, and R-F-M codes represent them. Thus, we first convert these quantities to scores based on the quintiles in the target dataset. This step of the RFM scoring was computed as follows.

1. Create an aggregated variable named *Total* by multiplying *Quantity* with *Price*, which gives the total amount of money spent per product/item in each transaction.

Table 3

Variables in the RFM scores and data (5,878 instances).

Variable name	Data type	Description
customer_id	Nominal	Corresponding to each distinct customer id
rfm_score	Nominal	Combination of recency, frequency, and monetary scores
transaction_count	Numeric	Frequency of purchase per customer
recency_days	Numeric	Recency in day
amount	Numeric	Monetary or total amount spent per customer
recency_score	Numeric	Recency scores [1, 5]
frequency_score	Numeric	Frequency scores [1, 5]
monetary_score	Numeric	Monetary scores [1, 5]

2. Create three essential aggregated variables *recency_days* (recency), *transaction_count* (frequency), and *amount* (monetary) to calculate the values of these variables for every customer:
 - Recency for variable *recency_days*: the difference between the analysis date and the most recent date that the customer has shopped in the store. The analysis date has been taken as the maximum date available for the variable *InvoiceDate*.
 - Frequency for variable *transaction_count*: The number of transactions performed by every customer.
 - Monetary for variable *amount*: Total money spent by every customer in the store.
3. Calculate RFM scores. All the values for frequency and monetary in the first quintile are given 1 for *frequency_score* and *monetary_score*, given 2 for the second quintile values, and so on. For recency, a more recent customer will have less recency value than a customer who has not shopped in a while. Therefore, the recency values in the first quintile are given 5 for *recency_score*, 4 for the second quintile values, and so on.
4. Create a variable named *rfm_score* by combining values of variables: *recency_score*, *frequency_score*, and *monetary_score*, respectively.

The result from data preparation with RFM analysis consists of 5,878 instances summarized in Table 3. This prepared dataset is provided to build the knowledge structure in the next step.

4.2. FCA knowledge structure

This step uses RFM scores as the data to generate a formal context. We first transform the RFM scores (with three attributes and five levels in each) into a formal context containing binary data using 15 attributes to analyze and discover the latent concepts addressed in the customers and their relationships. Afterwards, we create the formal concepts of RFM scores and then create the FCA lattice following equation (1) of the theory in subsection 2.3.

The hierarchical concept of this lattice will be built to provide a knowledge base of the proposed system. At the same time, the support values of the concepts will be computed to determine the implication association rules and their confidences. To construct the less complex hierarchical concept lattice with a high degree of visualization, we determine only the concepts whose number of objects is larger than a certain threshold.

From the hierarchical concept lattice, the customer segmentation is derived from the sublattice that considers the top view of this lattice. The relationship of customer behavior in each segmentation is derived from the implication rules following equations (2) – (3) of the theory in subsection 2.3. Those data will be prepared as described in the following subsection.

4.3. Business implication deployment

This subsection begins with experimenting with traditional types of clustering, namely K-means and agglomerative hierarchical clustering, by using RFM data and scores dataset.

As is well-known, the K-means clustering algorithm is very sensitive to outliers (data anomalies) or variables of incomparable scales or magnitudes. To find the best parameters for the K-means model, we proceeded with the following steps:

1. Removing the outliers: Some customers show unusual behavior in almost all businesses. Therefore, we had to remove these data for high-quality data analysis and apply machine learning methods. The critical point is that we used Interquartile Ranges (IQR) to remove the outliers. To calculate the IQR, the dataset is divided into quartiles or four rank-ordered even parts via linear interpolation. These quartiles are denoted by Q1 (also called the lower quartile), Q2 (the median), and Q3 (also called the upper quartile). Thus, we used the formula (4) below to find the threshold for our data's upper and lower bound in recency, frequency, and monetary features.

$$\begin{aligned} \text{IQR} &= Q3 - Q1 \\ \text{lower bound} &= Q1 - 1.5 \times \text{IQR} \\ \text{upper bound} &= Q3 + 1.5 \times \text{IQR} \end{aligned} \quad (4)$$

Accordingly, we removed the values below the lower or upper bound in each recency, frequency, and monetary. Finally, there were only 5,633 customers left by this screening.

2. Solving the Long Tail problem in recency, frequency, and monetary: The histograms of RFM values can be used to support our clustering step, but it leads to problems. Thus, we applied the Log Transformation to change the distribution of our data from a Long Tailed shape in histograms of RFM values to a more Normal distribution.
3. Scaling the range of frequency and monetary to have appropriate data for clustering methods: Since the range of values in our frequency was utterly different from monetary values. We had to apply normalization to both frequency and monetary data. We used Max-Min Scaling, which helped us scale our data to an appropriate range for clustering.
4. Finding the best Number of Segments (best K for K-means)

The above K-means clustering process results will be compared with our work. Unlike K-means, hierarchical clustering can produce a result without assigning initial values. In this work, we also used agglomerative hierarchical clustering to compare with our approach. Usually, the hierarchical clustering method is the most intuitive way of grouping data (Dolnicar et al. (2018b)). The hierarchical clustering applied Ward's method for similarity measures. However, the number of targeted clusters is a crucial parameter that should be determined before applying both the K-means and the hierarchical clustering algorithm. The Silhouette index and Davies-Boulbin index were chosen to assess the performances of clustering models. A higher Silhouette index or a lower Davies-Bouldin index is preferred for an optimal cluster number.

After the traditional clustering experimental results, the customer segmentation will be discussed by creating a descriptive profile for each customer segment with different characteristics based on the results of the proposed hierarchical concept and knowledge reported in the previous section. Finally, to design and deliver differentiated marketing strategies, the segmentation solution with customer relationship management and marketing implications are recommended for each customer group.

5. Results and discussion

This section presents the first to final steps and discusses the three steps mentioned above. There are (1) exploratory data analysis and preprocessing, (2) FCA knowledge structure, and (3) business implication deployment.

After preparing the dataset, RFM values will be visualized in the first step, shown in Figure 2, 3, and 4. These figures depict the distribution of RFM values (*recency_days*, *transaction_count*, and *amount*). It shows that there are outliers in the RFM values. Moreover, the histograms of RFM values illustrated in Figure 5 clearly show that the RFM values are not normally distributed. For this reason, the Spearman correlation coefficient and the Kendall correlation coefficient are used to assess the nonlinear dependences between the variables of the RFM model. The correlation matrices illustrated in Figure 6 emphasize that there is a strong correlation between the variables Frequency and Monetary. The results of RFM data and scores consisting of 5,878 instances are provided for the next step.

The second step involves the FCA knowledge structure. We used the fcaR package (Cordero, Enciso, López-Rodríguez and Mora (2022)) to perform FCA with R. The result of FCA contains 208 concepts. The first 11 formal concepts are shown in Table 4, and some concepts' sublattices are shown in Figure 7. This table and figure show a group of customers with the same attributes (behaviors).

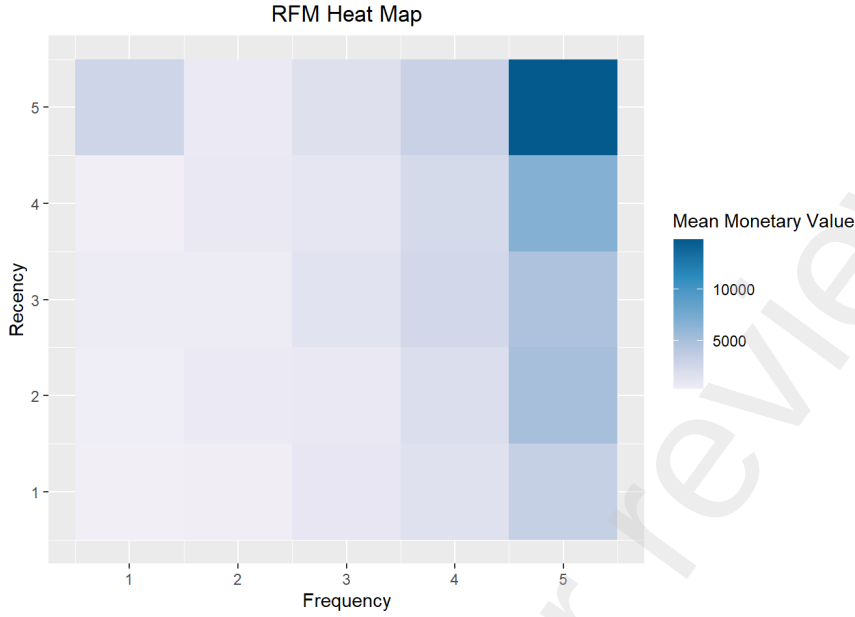


Figure 2: RFM score heatmap.

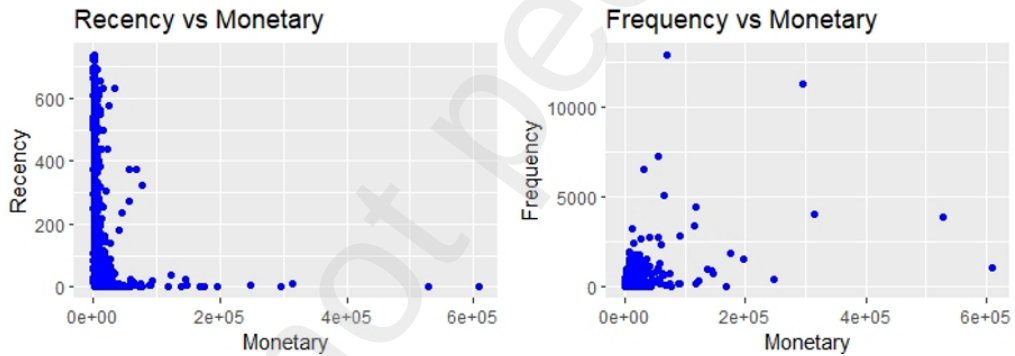


Figure 3: Distribution of Recency vs. Monetary and Frequency vs. Monetary values.

In addition, we extract association rules from the concept lattice. The general form is " $\langle N \rangle P = [C] \Rightarrow \langle N' \rangle C'$ ", where N is the number of objects satisfying the premise, P is a precondition, C is the confidence of association rules, N' is the number of objects meeting the premise, and C' is the conclusion. For example, the first implication in Table 5, i.e., $\langle 533 \rangle R5 M5 = [81\%] \Rightarrow \langle 434 \rangle F5$, means that there are 533 customers with a recency score of 5, and the monetary score of 5, and 434 customers among them have frequency score of 5. Thus, its confidence is 81%.

However, in this case, the number of formal concepts, with 208 concepts is excessively large. The concept lattices are ordered sets according to their extents or objects covered using the partial order relationship approach. Therefore, we use the top first concepts as the primary customer clusters. In this case, we use the first seven concepts, including concepts $\{M5\}$, $\{F5\}$, $\{M4\}$, $\{M3\}$, $\{M2\}$, and $\{M1\}$. Then, we considered the other concepts relating to these concepts.

The RFM scoring applies the quintile method by splitting data into five quintiles, and thus, in this case, we use only concepts whose support values are more than 0.04. We have a dataset of 5,878 customers. A concept is considered if this concept consists of at least $0.04 \times 5,878 = 235$ customers (20% of main clusters). Therefore, Figure 8 depicts the customer hierarchical concept.

For business implication deployment, the concept lattice in Figure 7 and the hierarchical concept in Figure 8 em-

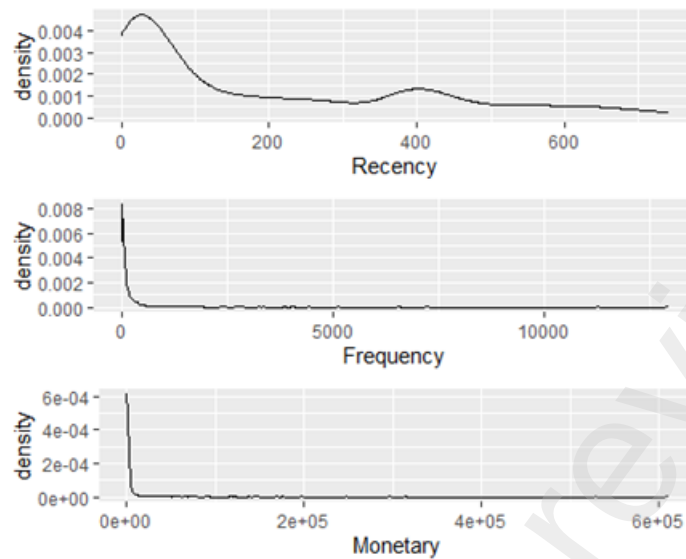


Figure 4: Distribution of RFM values.

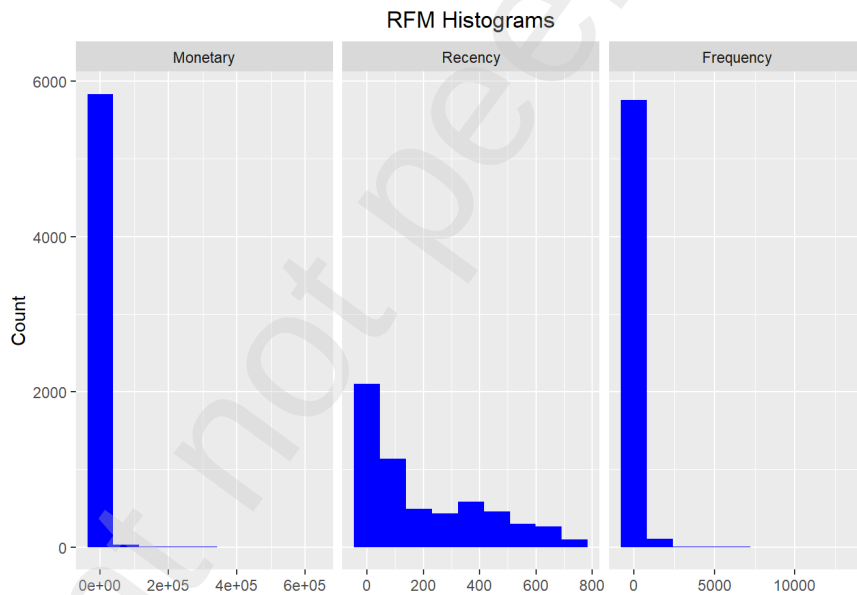


Figure 5: Histograms of RFM values.

phasize that the monetary score is the most significant variable in dividing customers into groups. Thus, we can simply split the customers with their monetary scores. Moreover, we also must monitor the customers having a frequency score of 5 in the highest quintile. The concept {F5} is associated with concept {F5, M5} and concept {F5, M4} belonging to the first and second quintile of monetary, respectively. The number of customers from these two concepts is 18% of all customers. Accordingly, we know that the more frequently the customer purchases, the more profit the company gains. The most frequent spenders with recent purchases are customers with the highest spending amount. In contrast, customers who made their transactions long ago and less frequently contributed low monetary scores.

From Table 7, the total purchase value of concept {M5} is 77.26% of the overall purchase value. It conforms to

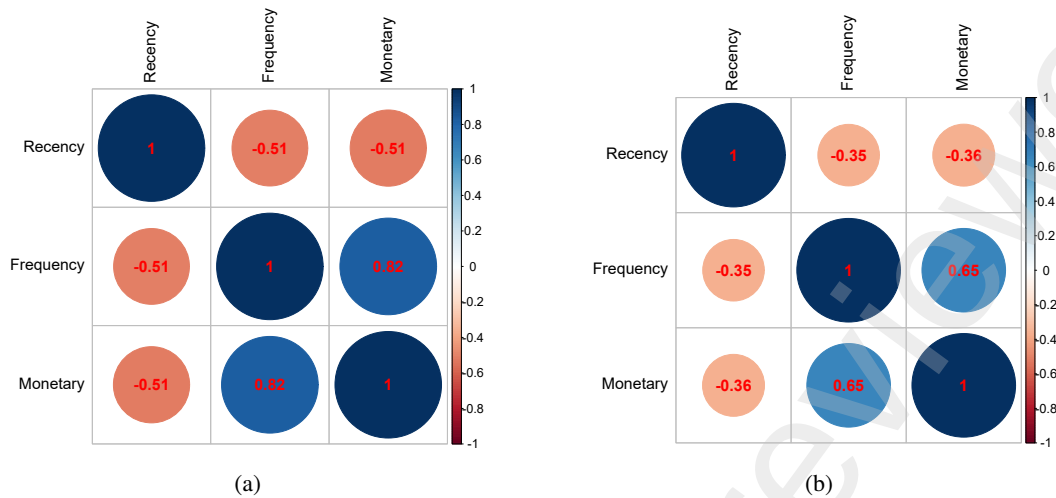


Figure 6: Matrices of correlation with (a) Spearman and (b) Kendall coefficients between the variables Recency, Frequency, and Monetary

Table 4

The first 11 formal concepts from all 208 concepts.

Concept#	Attributes	Support values	Number of customers
1	{}	1.000000	5878
2	{M5}	0.200068	1176
3	{M4}	0.199728	1174
4	{M3}	0.199388	1172
5	{M2}	0.199898	1175
6	{M1}	0.200919	1181
7	{F5}	0.199898	1175
8	{F5, M5}	0.143246	842
9	{F5, M4}	0.043552	256
10	{F5, M3}	0.012589	74
11	{F5, M2}	0.000510	3

the Pareto principle (Craft and Leake (2002); Kim, Singh and Winer (2017)) that "80% of sales comes from 20% of customers". Moreover, Table 7 shows that customers in concept {F5} spent 66.62% of company purchase value.

The information given in Figure 8 and Table 6 is used for analyzing the characteristics of the clusters.

Cluster 1 ({M5}) consists of 1176 customers or 20% of the total customers. The 842 customers, or 72% of total customers in this cluster, have frequency scores in the top quintile (F5). Concept {F5, M5} is a sub-concept of this concept {M5}. Moreover, the 237 customers, or 20% of cluster members, are in the fourth quintile (F4) of frequency values. Concept {F4, M5} is also a sub-concept of this concept {M5}. Consequently, it demonstrates that most of the customers in this group purchased products most frequently. Regarding the recency scores, the 533 customers, or 45% of cluster members, have recency scores in the top quintile (R5). Concept {R5, M5} is a sub-concept of this concept/cluster. Furthermore, the 326 customers, or 28% of cluster members, are in the fourth quintile (R4) of recency values. For this reason, most customers in this cluster most recently bought products. Above all, the best customers with RFM scores 5-5-5, and 4-5-5 in sub-concept {R5, F5, M5} and {R4, F5, M5} include 675 customers or 14% of the whole population. We can categorize this group as customers having the highest monetary score.

Cluster 2 ({M4}) contains 1174 customers or 20% of the whole customers. Moreover, the customers in this segment with frequency scores of F5 and F4 compose 22% and 49% of the cluster members in sequence.

Cluster 3 ({M3}) includes 1172 members or 20% of the total population. The 49% of customers in this group have frequency scores of 3 (F3) and monetary scores of 3 (M3).

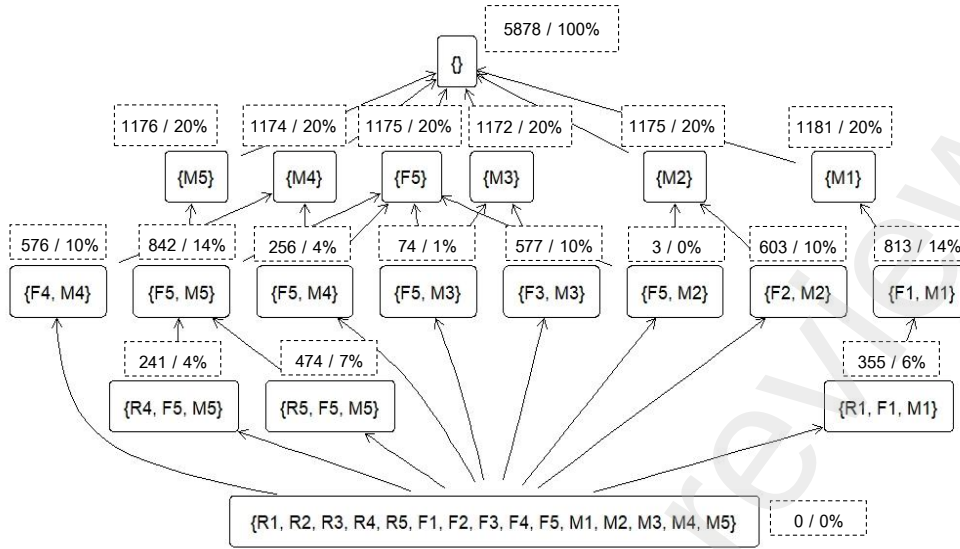


Figure 7: Sublattice of concept 1, 2, 3, 4, 5, 6, 7, 8, 14, 21, 28, 35, 43, 76, and 207 (non-specific object).

Table 5

List of association rules from concept lattice with confidence $\geq 60\%$.

No.	Rules	No.	Rules
1	$\langle 533 \rangle R5 \ M5 = [81\%] \Rightarrow \langle 434 \rangle F5$	10	$\langle 1178 \rangle M1 = [69\%] \Rightarrow \langle 813 \rangle F1$
2	$\langle 546 \rangle R5 \ F5 = [79\%] \Rightarrow \langle 434 \rangle M5$	11	$\langle 515 \rangle R1 \ M1 = [69\%] \Rightarrow \langle 355 \rangle F1$
3	$\langle 475 \rangle R1 \ F1 = [75\%] \Rightarrow \langle 355 \rangle M1$	12	$\langle 128 \rangle R4 \ M1 = [69\%] \Rightarrow \langle 88 \rangle F1$
4	$\langle 326 \rangle R4 \ M5 = [74\%] \Rightarrow \langle 241 \rangle F5$	13	$\langle 1189 \rangle F1 = [68\%] \Rightarrow \langle 813 \rangle M1$
5	$\langle 168 \rangle R3 \ M1 = [73\%] \Rightarrow \langle 122 \rangle F1$	14	$\langle 71 \rangle R5 \ M1 = [68\%] \Rightarrow \langle 48 \rangle F1$
6	$\langle 122 \rangle R4 \ F1 = [72\%] \Rightarrow \langle 88 \rangle M1$	15	$\langle 296 \rangle R2 \ M1 = [68\%] \Rightarrow \langle 200 \rangle F1$
7	$\langle 1175 \rangle F5 = [72\%] \Rightarrow \langle 842 \rangle M5$	16	$\langle 305 \rangle R2 \ F1 = [66\%] \Rightarrow \langle 200 \rangle M1$
8	$\langle 1176 \rangle M5 = [72\%] \Rightarrow \langle 842 \rangle F5$	17	$\langle 182 \rangle R3 \ F5 = [62\%] \Rightarrow \langle 113 \rangle M5$
9	$\langle 339 \rangle R4 \ F5 = [71\%] \Rightarrow \langle 241 \rangle M5$	18	$\langle 80 \rangle R5 \ F1 = [60\%] \Rightarrow \langle 48 \rangle M1$

Cluster 4 ($\{M2\}$) includes 1175 members or 20% of population. Most customers in this group, with 51% of cluster members, have frequency scores of 2 (F2) and monetary scores of 2 (M2).

Cluster 5 ($\{M1\}$) includes 1181 members or 20% of population. The customers in this group are the lost customers or the lowest potential customers. Most customers in this group, with 69% of cluster members, have frequency scores of 1 (F1) and monetary scores of 1 (M1). In addition, 44% of cluster members have recency scores of 1 (R1) and monetary scores of 1 (M1). The number of worst customers with RFM scores of 111 in sub-concept $\{R1, F1, M1\}$ is 33% of all customers in this group.

Cluster 6 ($\{F5\}$) comprises 1175 members or 20% of total customers. Most customers in this group, with 72% of group members, have frequency scores of 5 (F5) and monetary scores of 5 (M5). In addition, 46% of customers in this cluster have recency scores of 5 (R5) and frequency scores of 5 (F5). The customers in this cluster most frequently and recently purchased products.

To compare and discuss traditional K-means clustering and hierarchical clustering, the results of using the Silhouette index and Davies-Boulbin index to assess the performance of the traditional clustering model are shown in Figures 9 and 10, respectively. We found that 4 – 6 clusters is an appropriate choice because then they are meaningful and valuable in business implications. The number of customer segments should not be too large because then it will be challenging to interpret and design marketing strategies. In addition, 4 – 5 groups is the choice in most of the

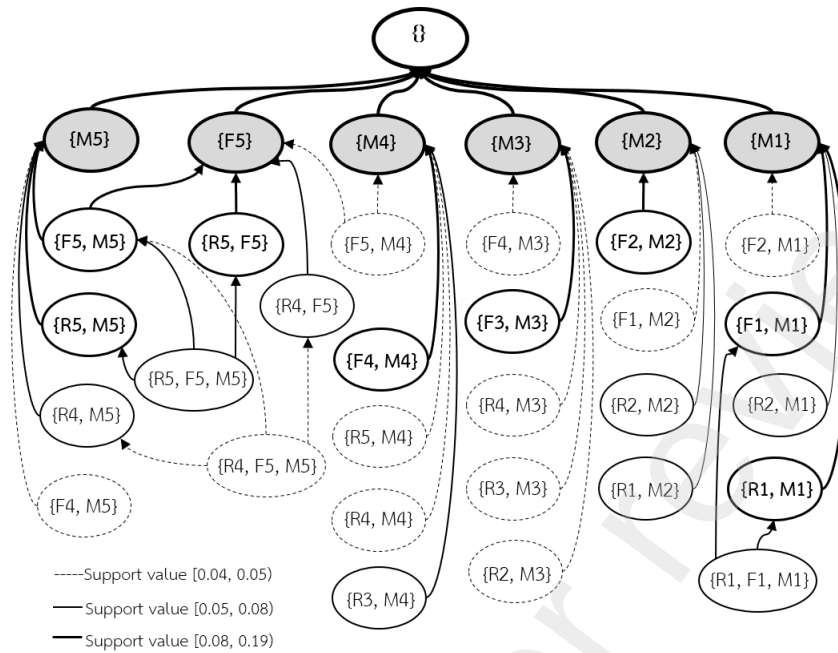


Figure 8: RFM-based customer hierarchical concept.

Table 6

Description of the concepts related to Figure 8.

Concepts	Support values	Number of customers	Concepts	Support values	Number of customers
{M5}	0.200068	1176	{R5, M4}	0.048826	287
{M4}	0.199728	1174	{R5, F5}	0.092889	546
{M3}	0.199388	1172	{R5, F5, M5}	0.073835	434
{M2}	0.199898	1175	{R4, M5}	0.055461	326
{M1}	0.200919	1181	{R4, M4}	0.048146	283
{F5}	0.199898	1175	{R4, M3}	0.041170	242
{F5, M5}	0.143246	842	{R4, F5}	0.057673	339
{F5, M4}	0.043552	256	{R4, F5, M5}	0.041000	241
{F4, M5}	0.040320	237	{R3, M4}	0.054440	320
{F4, M4}	0.097993	576	{R3, M3}	0.044573	262
{F4, M3}	0.040490	238	{R2, M3}	0.047976	282
{F3, M3}	0.098163	577	{R2, M2}	0.057503	338
{F2, M2}	0.102586	603	{R2, M1}	0.050357	296
{F2, M1}	0.044573	262	{R1, M2}	0.056822	334
{F1, M1}	0.138312	813	{R1, M1}	0.087955	517
{R5, M5}	0.090677	533	{R1, F1, M1}	0.060395	355

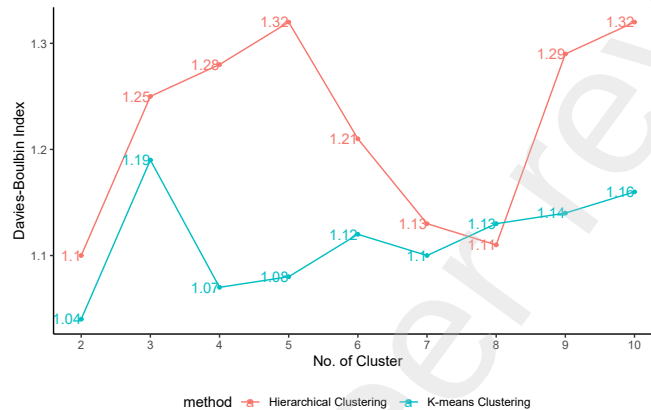
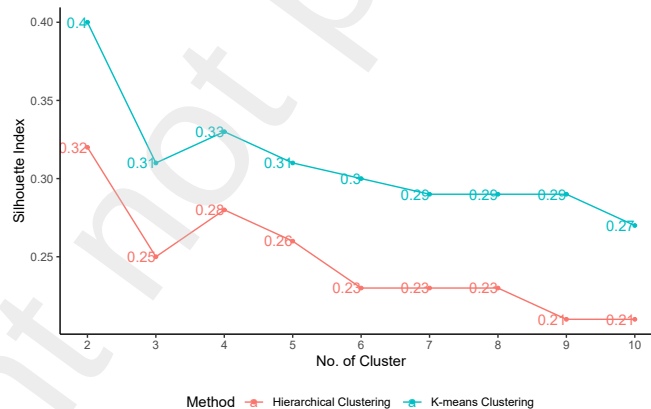
prior works (Ernawati et al. (2021)), and we need to compare the results with those studies. Therefore, we divided the customers into six main concepts based on FCA. The cluster profiles and results summarizing traditional K-means clustering and hierarchical clustering are provided in Figures 11 and 12, and Tables 8 and 9.

The experimental results show that FCA is not sensitive to outliers. Thus, it is not necessary to eliminate the outliers. On the other hand, the K-means algorithm needs outlier management and data normalization. The proposed technique uses the RFM quintile method to create RFM scores. Then, the formal context is created using all customers' RFM scores. This new methodology combines RFM analysis and FCA; therefore, the relationships can be visualized in a Hasse diagram for the hierarchical concept. K-means clustering gives only completely separate

Table 7

Customer segmentation with FCA and total purchase value by cluster.

Cluster/Concept	Total Purchase Value	% of Total Purchase Value	% of Customer Number
{M1}	194,849.11	1.10%	20%
{M2}	507,220.44	2.86%	20%
{M3}	1,061,419.95	5.98%	20%
{M4}	2,271,513.61	12.80%	20%
{M5}	13,708,426.06	77.26%	20%
{F5}	11,821,362.94	66.62%	20%

**Figure 9:** Davies-Boulbin index of hierarchical and K-means clustering at different cluster numbers.**Figure 10:** Silhouette index of hierarchical and K-means clustering at different cluster numbers.

customer groups without their relationships and implicit knowledge. FCA also has a strong mathematical theory as support. Thus, the results of FCA can be assessed using mathematical theory. Unlike K-means, whose results are unstructured non-overlapping clusters, this new approach mainly creates hierarchical overlapping clusters. Therefore, a customer possibly (usually) belongs to more than one hierarchical concept or cluster. For this reason, marketers have increased opportunities to deliver the customers an appropriate matched marketing strategy to increase customer retention and satisfaction. Above all, this new approach provides ease of processing and understanding of the results. The comparison summary is shown in Table 10.

To deploy in marketing strategy and recommendations, the total purchase value of the most valuable customers is

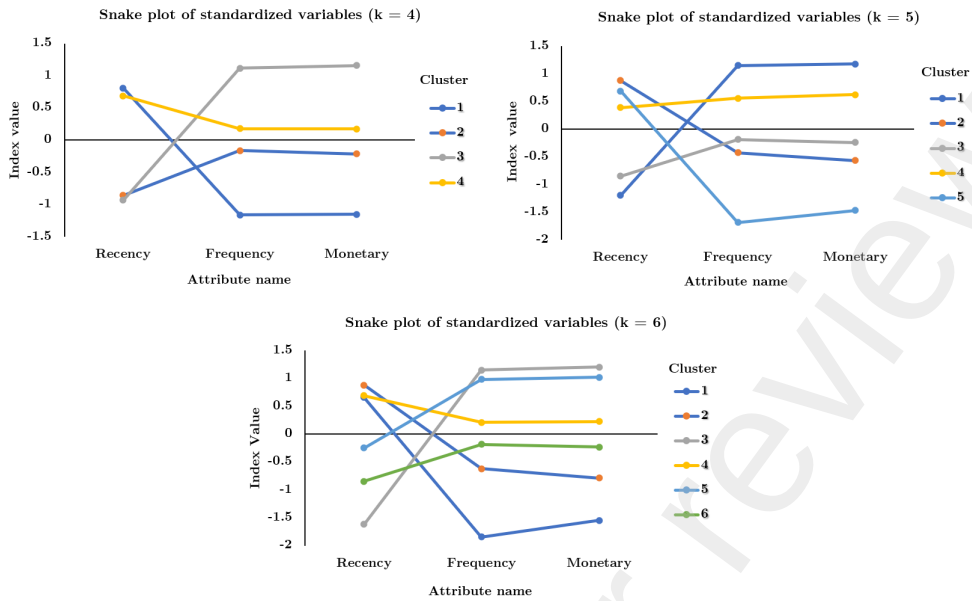


Figure 11: Snake plots of standardized variables of K-means clustering output.

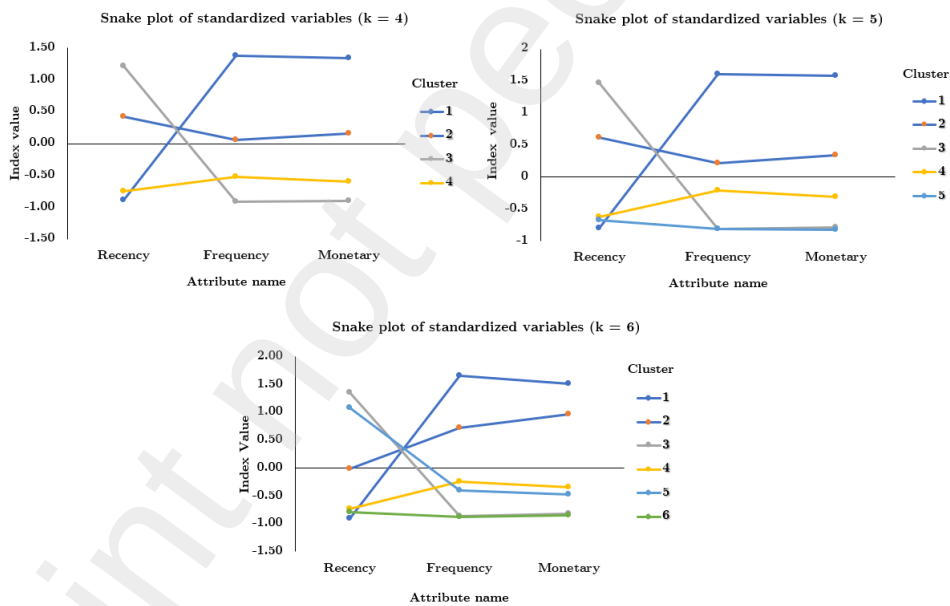


Figure 12: Snake plots of standardized variables of hierarchical clustering output.

almost 80 percent of total sales during the data accumulation period. On the other hand, the company gains only 1 percent of the total purchases from the worst customer group. The best customers are high potential targets for new products. They will help the company to promote its products and brand. Therefore, the company should reward these customers and maintain close relationships to increase revenue. In addition, the company should understand why the worst customers or lost customers did not buy the products anymore. For customers with average frequency and good total sales, the company should offer promotions and recommend products for upselling. The company should provide special offers to increase customer visits, leading to more purchases.

Table 8

K-means clustering output summary related to Figure 11.

k = 4					
Cluster#	Recency (Mean)	Frequency (Mean)	Monetary (Mean)	Count	Description
1	High (394.26)	Low (13.66)	Low (248.20)	1472	Lost customers
2	Low (32.91)	Medium (48.10)	Medium (765.57)	1121	Potential loyalists
3	Low (37.75)	High (251.59)	High (4205.91)	1432	Champions
4	High (311.10)	Medium (75.13)	Medium (1299.57)	1608	Hibernating customers
k = 5					
Cluster#	Recency (Mean)	Frequency (Mean)	Monetary (Mean)	Count	Description
1	Low (20.95)	High (264.09)	High (4369.32)	1141	Champions
2	High (399.47)	Medium Low (30.88)	Medium Low (480.56)	1450	Hibernating customers
3	Low (32.73)	Medium (46.31)	Medium (737.69)	1061	Potential loyalists
4	Medium High (216.01)	Medium High (122.99)	Medium High (2173.25)	1263	At risk
5	High (363.34)	Low (6.31)	Low (176.06)	718	Lost customers
k = 6					
Cluster#	Recency (Mean)	Frequency (Mean)	Monetary (Mean)	Count	Description
1	High (360.09)	Low (5.00)	Low (168.14)	565	Lost customers
2	High (407.47)	Medium Low (23.20)	Medium Low (349.47)	1238	Hibernating customers
3	Low (8.91)	High (273.09)	High (4631.04)	683	Champions
4	High (308.96)	Medium High (73.98)	Medium High (1279.63)	1105	At risk
5	Medium (82.73)	High (208.63)	High (3479.89)	990	Loyal customers
6	Medium Low (32.58)	Medium (46.34)	Medium (747.53)	1052	Potential loyalists

6. Conclusion

This research proposes a new effective clustering algorithm using the advantages of FCA to build a knowledge representation. This model combines the RFM model with FCA. Thus, the construction contains both implicit and explicit knowledge. Explicit knowledge shows cluster visualized information represented in the hierarchical structure model, while implicit knowledge is embedded in the structure with its implication properties. Thus, the knowledge structure from FCA reveals relationships among data points and easily understood results. Afterwards, the proposed model was compared with K-means clustering and hierarchical clustering using the online retail II dataset from the UCI Machine Learning Repository. In conclusion, the proposed method provides enough and appropriate information for marketers to perceive the value of the clustering results for creating practical marketing strategies in real-world business. This approach offers marketers both customer segmentation and relationships in customer data simultaneously.

The advantage of the RFM model is the use of a very small number of variables (only three variables) to reduce the complexity of the model. However, we suggest adding new variables to the RFM model to increase the accuracy and gain more information in future studies. For example, RFMT was introduced in Zhou et al. (2021) by adding a variable, interpurchase time (T), for more effective customer segmentation. Moreover, there are many other models with added variables, such as LRFM (Chang and Tsay (2004)), RFMTC (Yeh, Yang and Ting (2009)), and RFMD (Noori (2015)).

In addition, we will modify and improve this model by representing RFM values in a non-binary formal context in future studies. Then, FCA will be considered and compared with the results from this alternative model.

This proposed method can also be applied to other businesses than online retail. Therefore, we recommend employing it in the context of other industries in future research.

Table 9

Hierarchical clustering output summary related to Figure 12.

k = 4					
Cluster#	Recency (Mean)	Frequency (Mean)	Monetary (Mean)	Count	Description
1	Low (14.01)	High (265.24)	High (4302.61)	913	Champions
2	Medium High (258.88)	Medium (122.63)	Medium (2223.53)	1594	Hibernating customers
3	High (406.21)	Low (17.78)	Low (346.06)	1688	Lost customers
4	Low (42.029)	Medium Low (60.81)	Medium Low (889.69)	1438	Potential loyalists
k = 5					
Cluster#	Recency (Mean)	Frequency (Mean)	Monetary (Mean)	Count	Description
1	Low (14.01)	High (265.24)	High (4302.61)	913	Champions
2	Medium High (258.88)	Medium (122.63)	Medium (2223.53)	1594	Hibernating customers
3	High (406.21)	Low (17.78)	Low (346.06)	1688	Lost customers
4	Low (44.75)	Medium Low (78.76)	Medium Low (1137.65)	1020	Potential loyalists
5	Low (35.39)	Low (17.03)	Low (284.63)	418	New customers
k = 6					
Cluster#	Recency (Mean)	Frequency (Mean)	Monetary (Mean)	Count	Description
1	Low (14.01)	High (265.24)	High (4302.61)	913	Champions
2	Medium (169.74)	Medium High (173.95)	Medium High (3371.74)	848	Loyal customers
3	High (406.21)	Low (17.78)	Low (346.06)	1688	Lost customers
4	Low (44.75)	Medium Low (78.76)	Medium Low (1137.65)	1020	Potential loyalists
5	High (360.20)	Medium Low (64.29)	Medium Low (918.33)	746	Hibernating customers
6	Low (35.39)	Low (17.03)	Low (284.63)	418	New customers

Table 10

FCA-based and K-means clustering comparison.

Issues	FCA	K-means	Hierarchical
Outlier	Not sensitive to outliers	Sensitive to outliers and outlier management needed	Sensitive to outliers leading to imbalanced clustering results
Data normalization	RFM quintile method used	Data normalization needed	Data normalization needed
Relationship information among segments	Relationship demonstration with Hasse diagram and hierarchical concept	Complete separation of customer groups	Complete separation of customer groups with hierarchically structured dendrogram
Mathematical theory support	Strong support	Cluster centroid initialization sensitivity	Imbalanced clustering results.
Implicit Knowledge	Included	Not included	Not included
Clustering result	Hierarchical overlapping clusters	Unstructured non-overlapping clusters	Hierarchical structured non-overlapping clusters
Preprocessing	Less	More	More
Visualization	Easy	More difficult	More difficult
Clustering type	Soft clustering	Hard clustering	Hard clustering

7. Acknowledgements

This work was supported by the Digital Science for Economy, Society, Human Resources Innovative Development and Environment project funded by Reinventing Universities Research Institutes under grant no. 2046735, Ministry of Higher Education, Science, Research and Innovation, Thailand.

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